Systems Analysis and Control

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Lecture 8: Response Characteristics

Overview

In this Lecture, you will learn:

The Effects of Feedback on Dynamic Response

- Changes in Transfer Function
- Stability
- Impact of Poles on Dynamic Response

Real Poles

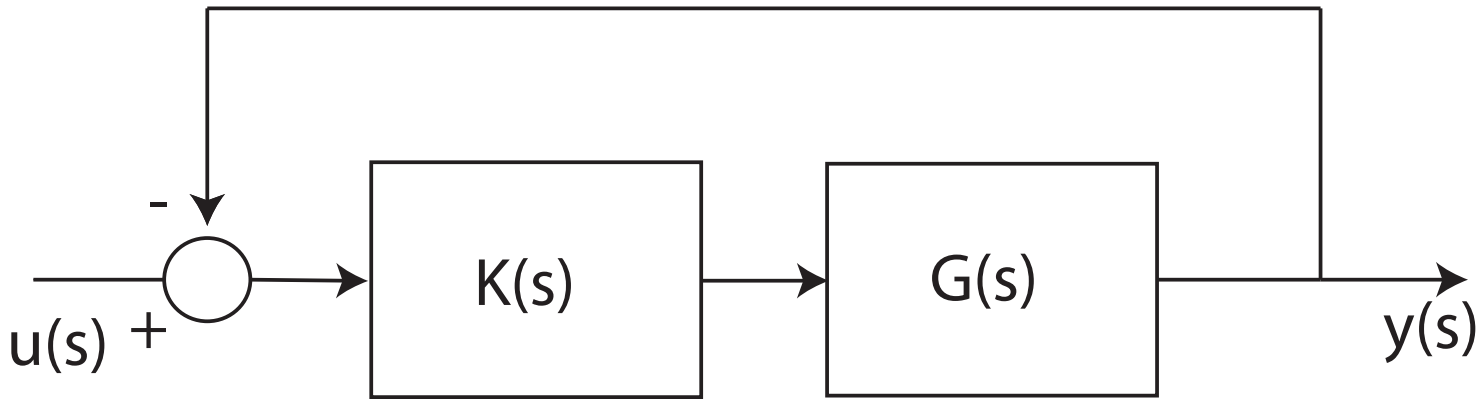
- Steady-State Error
- Rise Time
- Settling Time

Complex Poles

- Complex Pole Locations
- Damped/Natural Frequency
- Damping and Damping Ratio

Recall: The Effect of Feedback Control

The Upper Feedback Interconnection



Feedback:

- **Controller:** $\hat{u}(s) = K(s)(\hat{u}(s) - \hat{y}(s))$
- **Plant:** $\hat{y}(s) = \hat{G}(s)\hat{u}(s)$

The output signal is $\hat{y}(s)$,

$$\hat{y}(s) = \frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s)$$

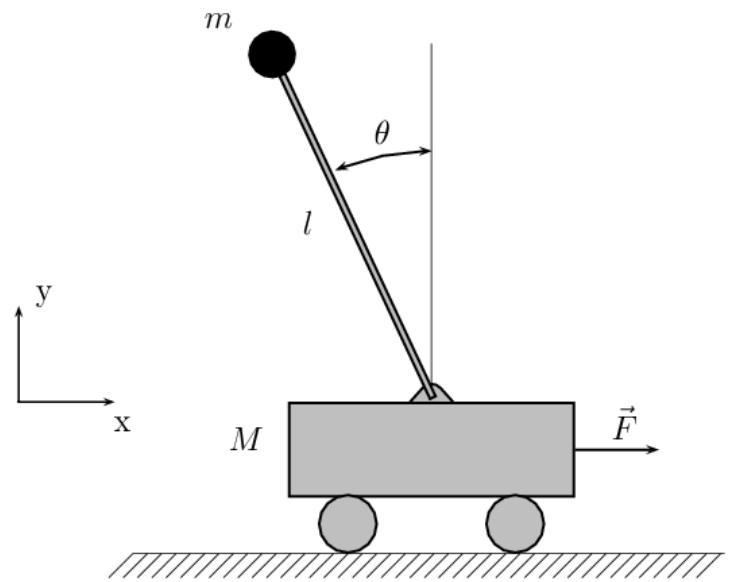
Question: How does $K(s)$ change the *behaviour* of the system?

Recall: Feedback and Inverted Pendulum Model

Open Loop Transfer Function

$$\hat{G}(s) = \frac{1}{Js^2 - \frac{Mgl}{2}}$$

Controller: Static Gain: $\hat{K}(s) = K$
Input: Impulse: $\hat{u}(s) = 1$.



Closed Loop: Lower Feedback Interconnection

$$\hat{y}(s) = \frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s) = \frac{\frac{K}{Js^2 - \frac{Mgl}{2}}}{1 + \frac{K}{Js^2 - \frac{Mgl}{2}}} = \frac{K}{Js^2 - \frac{Mgl}{2} + K}$$

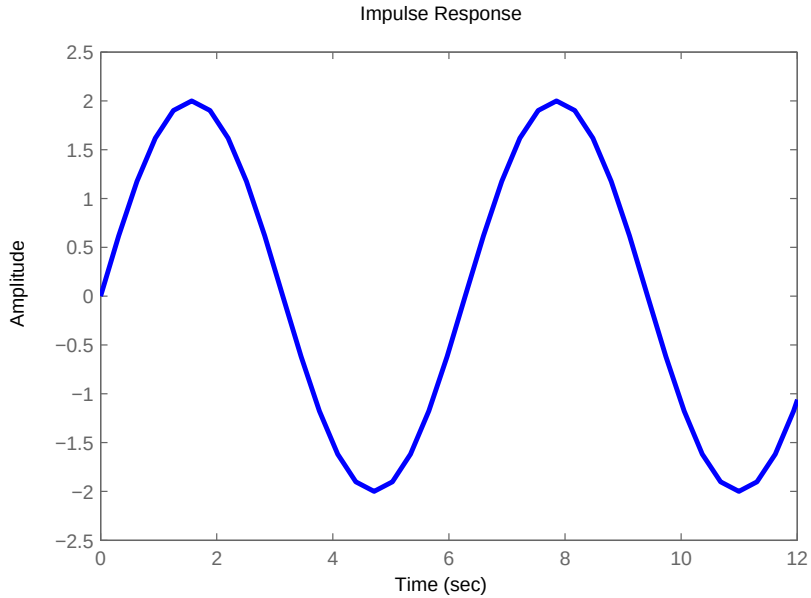
Recall: Feedback and the Inverted Pendulum Model

Closed Loop Impulse Response:
Lower Feedback Interconnection

$$\hat{y}(s) = \frac{\frac{K}{J}}{s^2 - \frac{Mgl}{2J} + \frac{K}{J}}$$

Recall:

- If $K \geq \frac{Mgl}{2}$: Infinite Oscillations
- If $K < \frac{Mgl}{2}$: Exponential Growth

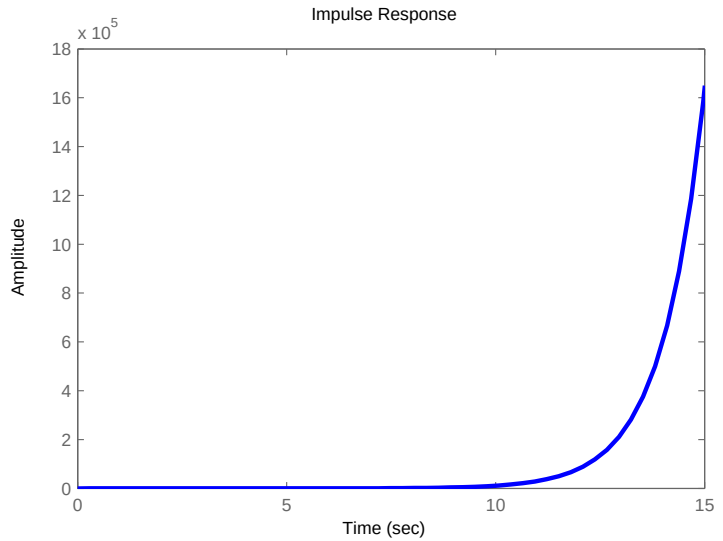


Open Loop: $K = 0$

$$\hat{y}(s)$$

$$= \frac{1}{J}\sqrt{\frac{2J}{Mgl}}\left(\frac{1}{s - \sqrt{\frac{Mgl}{2J}}} - \frac{1}{s + \sqrt{\frac{Mgl}{2J}}}\right)$$

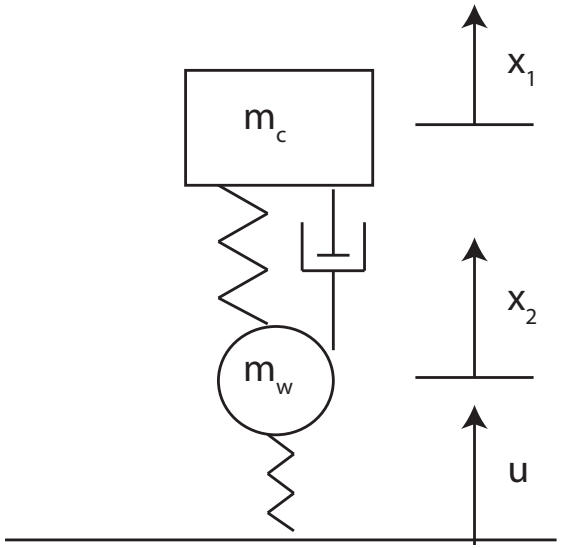
Unstable



Feedback and the Suspension System

Open Loop Transfer Function:
Set $m_c = m_w = g = c = K_1 = K_2 = 1$.

$$\hat{G}(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$



Controller: Static Gain: $\hat{K}(s) = k$

Closed Loop: Lower Feedback Interconnection:

$$\hat{y}(s) = \frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s)$$

$$= \frac{k(s^2 + s + 1)}{s^4 + 2s^3 + (3+k)s^2 + (1+k)s + (1+k)}$$

Feedback and the Suspension System

Effect of changing the Feedback, k

Closed Loop Step Response:

$$\hat{y}(s) = \frac{k(s^2+s+1)}{s^4+2s^3+(3+k)s^2+(1+k)s+(1+k)}\frac{1}{s}$$

High k:

- Overshot the target
- Quick Response
- Closer to $y(15) = u = 1$

Low k:

- Slow Response
- No overshoot
- Final value is farther from 1.

Questions:

- Which Traits are important?
- How to predict the behaviour?

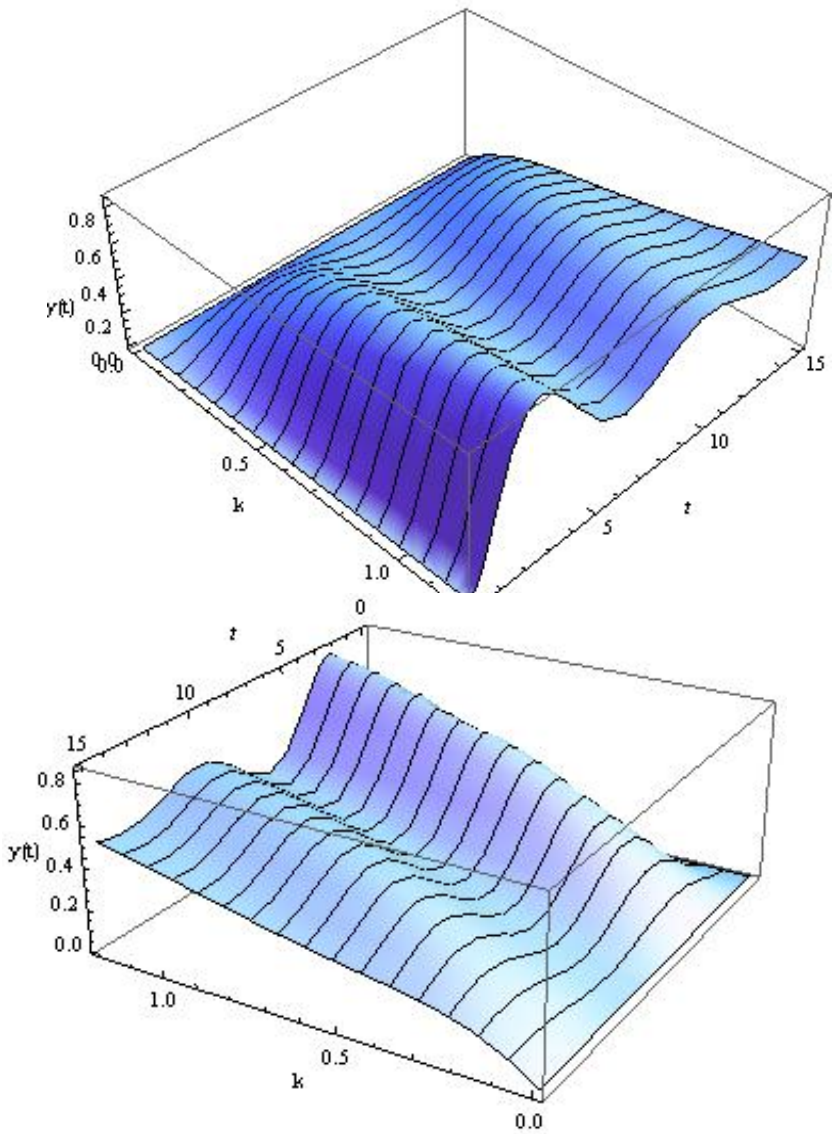


Figure: Step Response for different k

4 Important System Properties

We will **quantify** system performance using 4 properties

1. **Stability:** The most basic property
   - needed for all other definitions to be valid!
2. **Steady-State Error:** How well does output track input over long times
   - Does $\lim_{t\to\infty} y(t) = u(t)$?
3. **Rise Time:** How quickly does output track input over short times?
4. **Settling Time:** How quickly does output track input over long times?
5. **Peak Time and Percent Overshoot:** How big are the oscillations?
   - Only valid for complex poles – See part 2

Approach: Properties 1,3-5 are defined solely in terms of the pole locations

- We ignore the zeros (except for SS error)!

WARNING: Properties 3-4 are defined *differently* for real and complex poles.

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Formally, we are defining stability as

Definition 1.

A system, G is **Stable** if there exists a $K > 0$ such that

$$\|Gu\|_{L_2} \leq K\|u\|_{L_2}$$

Note: Although this is the true definition for systems defined by transfer functions, it is rarely used.

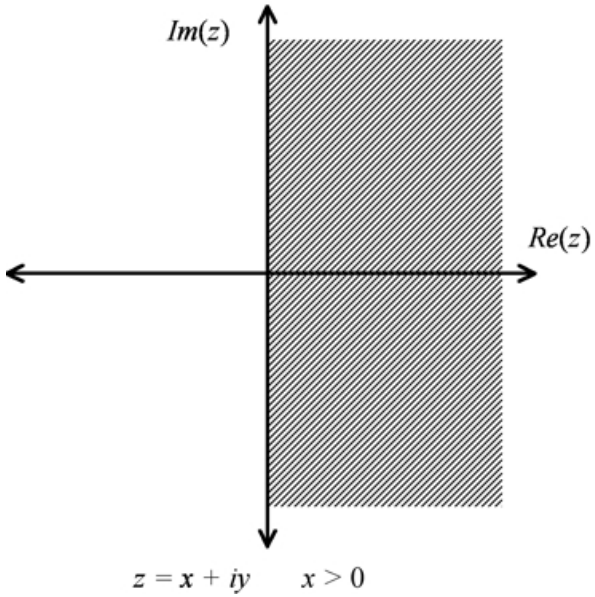
- Bounded input means bounded output.
- Stable means $y(t) \to 0$ when $u(t) \to 0$.

Property 1: Stability Depends on Pole Locations

Definition 2.

The **Closed Right Half-Plane**, $CRHP$ is the set of complex numbers with non-negative real part.

$$\{s \in \mathbb{C} : \text{Real}(s) \geq 0\}$$



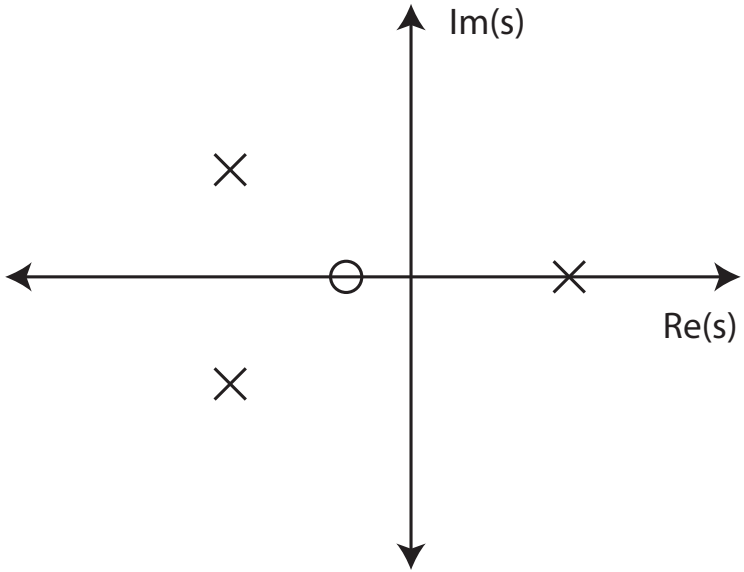


Figure: Even one unstable pole makes $\hat{G}$ unstable

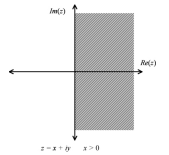
Theorem 3.

A system G , with transfer function $\hat{G}(s) = \frac{n(s)}{d(s)}$ is **Stable** *if it has* **NO** *poles in the Closed Right Half-Plane.*

- Check stability by checking poles.
- $s = x$ is a pole if $d(s) = 0$
- $s = o$ is a zero is $n(s) = 0$

Property 1: Stability Depends on Pole Locations

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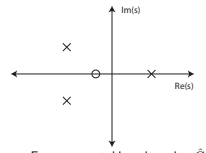


Figure: Even one unstable pole makes $\hat{G}$ unstable

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- Check stability by checking poles.
- $s = x$ is a pole if $d(s) = 0$
- $s = o$ is a zero is $n(s) = 0$

Recall if

$$y(s) = \frac{n(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)} = \frac{r_1}{s-p_1} + \ldots + \frac{r_n}{s-p_n}$$

Poles Dominate the Motion: Because the signal is the sum of poles, The inverse Laplace has the form

$$y(t) = r_1 e^{p_1 t} + \ldots + r_n e^{p_n t}$$

- If $p_i = x + \imath y$, then $r_i e^{p_i t} = r_i e^{(x+\imath y)t} = r_i e^{xt} e^{\imath y t}$
- So if $x > 0$, $y(t) \to \infty$

Property 2: Steady-State Error

Steady-State Error: How well does output track input over long times

- Does $\lim_{t\to\infty} y(t) = u(t)$?

Definition 4.

Steady-State Error for a stable system is the final difference between input and output.

$$e_{ss} = \lim_{t\to\infty} y(t) - u(t)$$

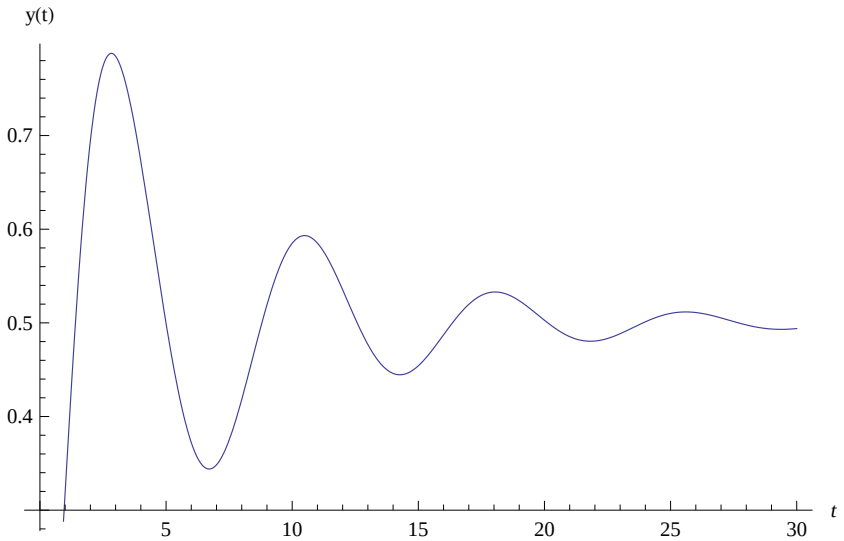
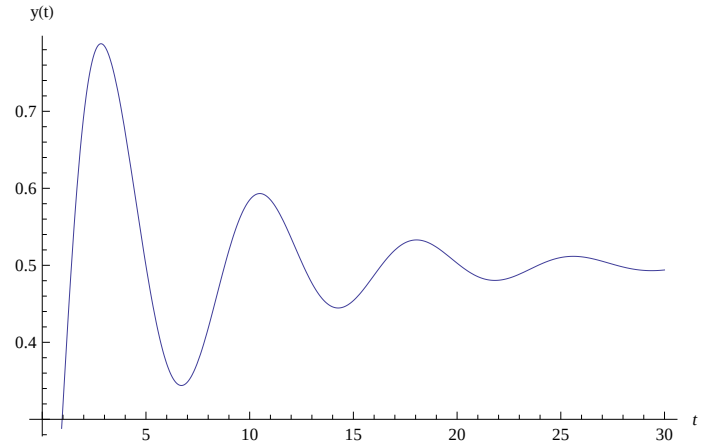


Figure: Suspension Response for $k = 1$

- Usually measured using the step or ramp response.
  - Since $u(t) = 1$, $e_{ss} = \lim_{t\to\infty} y(t) - 1$
- For step response its also the *percentage* error
  - $e_{ss} = \lim_{t\to\infty} \frac{y(t)-u(t)}{u(t)}$

Predicting Steady-State Value Using the Residue



Recall: For any system G, by partial fraction expansion:

$$\hat{y}(s) = \hat{G}(s)\frac{1}{s} = \frac{r_0}{s} + \frac{r_1}{s-p_1} + \ldots + \frac{r_n}{s-p_n}$$

So

$$y(t) = r_0\mathbf{1}(t) + r_1 e^{p_1 t} + \ldots + r_n e^{p_n t}$$

and if all the poles, p_i are stable, this means

$$\lim_{t\to\infty} y(t) = r_0$$

and hence

$$e_{ss} = r_0 - 1$$

The Final Value Theorem

The steady-state error is given by r_0.

$$e_{ss} = r_0 - 1$$

Recall: The residue at $s = 0$ is r_0 and is found as

$$r_0 = \hat{G}(s)|_{s=0} = \lim_{s \to 0} \hat{G}(s)$$

Thus the steady-state error is

$$e_{ss} = \lim_{s \to 0} \hat{G}(s) - 1$$

This can be generalized to find the limit of any signal:

Theorem 5 (Final Value Theorem).

$$\lim_{t \to \infty} y(t) = \lim_{s \to 0} s\hat{y}(s)$$

- Assumes the limit exists (Doesn't work for sinusoids: e.g. $y(s) = \frac{1}{s^2+1}$)
- Can be used to find response to other inputs
  - Ramp, impulse, etc.

The Final Value Theorem for Systems in Lower Feedback

Lower Feedback Interconnection: Tracking Framework

$$\hat{y}(s) = \frac{G(s)K}{1+G(s)K}\hat{u}(s)$$

Error Response for Lower Feedback Interconnection:

$$\begin{aligned}\hat{e}(s) &= \hat{y}(s) - \hat{u}(s) \\ &= \frac{G(s)K}{1+G(s)K}\hat{u}(s) - \frac{1+G(s)K}{1+G(s)K}\hat{u}(s) = \frac{-1+G(s)K-G(s)K}{1+G(s)K}\hat{u}(s) \\ &= \frac{-1}{1+G(s)K}\hat{u}(s)\end{aligned}$$

So the steady-state error is

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s\hat{e}(s) = -\lim_{s\to 0}\frac{s}{1+G(s)K}\hat{u}(s)$$

Error in Step Response: If $\hat{u}(s) = \frac{1}{s}$, then

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0}\frac{-1}{1+G(s)K}$$

Conclusion: Increasing K always reduces steady-state error to step!

The Final Value Theorem for Systems in Upper Feedback

Upper Feedback Interconnection: Regulator Framework

$$\hat{y}(s) = \frac{G(s)}{1 + G(s)K}\hat{u}(s)$$

Error Response for Upper Feedback Interconnection:

$$\begin{aligned}\hat{e}(s) &= \hat{y}(s) - \hat{u}(s) \\ &= \frac{G(s)}{1 + G(s)K}\hat{u}(s) - \frac{1 + G(s)K}{1 + G(s)K}\hat{u}(s) = \frac{G(s) - 1 - G(s)K}{1 + G(s)K}\hat{u}(s) \\ &= \frac{G(s)(1 - K) - 1}{1 + G(s)K}\hat{u}(s)\end{aligned}$$

Error in Step Response: If $\hat{u}(s) = \frac{1}{s}$, then

$$\lim_{t \to \infty} e(t) = \lim_{s \to 0} \frac{G(s)(1 - K) - 1}{1 + G(s)K}$$

Conclusion: Increasing K doesn't help with step response!

Steady-State Error

Numerical Example

$$\hat{G}(s) = \frac{k(s^2 + s + 1)}{s^4 + 2s^3 + (3+k)s^2 + (1+k)s + (1+k)}$$

The steady-state response is

$$\begin{aligned} y_{ss} &= \lim_{s \to 0} s\hat{y}(s) = \lim_{s \to 0} \hat{G}(s) \\ &= \lim_{s \to 0} \frac{k(s^2 + s + 1)}{s^4 + 2s^3 + (3+k)s^2 + (1+k)s + (1+k)} \\ &= \frac{k}{1+k} \end{aligned}$$

The steady-state error is

$$\begin{aligned} e_{ss} &= y_{ss} - 1 = \frac{k}{1+k} - 1 \\ &= \frac{-1}{1+k} \end{aligned}$$

- When $k = 0$, $e_{ss} = -1$
- As $k \to \infty$, $e_{ss} = 0$

Ramp Response

For Systems in Lower Feedback

Lower Feedback Interconnection: Recall the steady-state error is

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s\hat{e}(s) = \lim_{s\to 0} \frac{s}{1+G(s)K}\hat{u}(s)$$

Recall the Ramp input: If $u(t) = t$, then $\hat{u}(s) = \frac{1}{s^2}$.

Error in Ramp Response: In this case,

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} -\frac{1}{s(1+G(s)K)}$$

Conclusion: $\lim_{t\to\infty} e(t) = -\infty$!

Caution: Steady-state error of $G(s)$ to a ramp is only finite if $G(s)$ has a $\frac{1}{s}$ term.

Dynamic Response Characteristics

Settling Time and Rise Time

From Partial Fractions Expansion, you know that motion is determined by the poles of the Transfer Function!

$$\frac{n(s)}{(s^2+as+b)(s-p_1)\cdots(s-p_n)}\hat{u}(s) = \frac{k_1 s + k_2}{s^2+as+b}\hat{u}(s) + \frac{r_1}{s-p_1}\hat{u}(s) + \cdots + \frac{r_n}{s-p_n}\hat{u}(s)$$

- Simplify the response by considering response of each real pole and complex pair of poles.

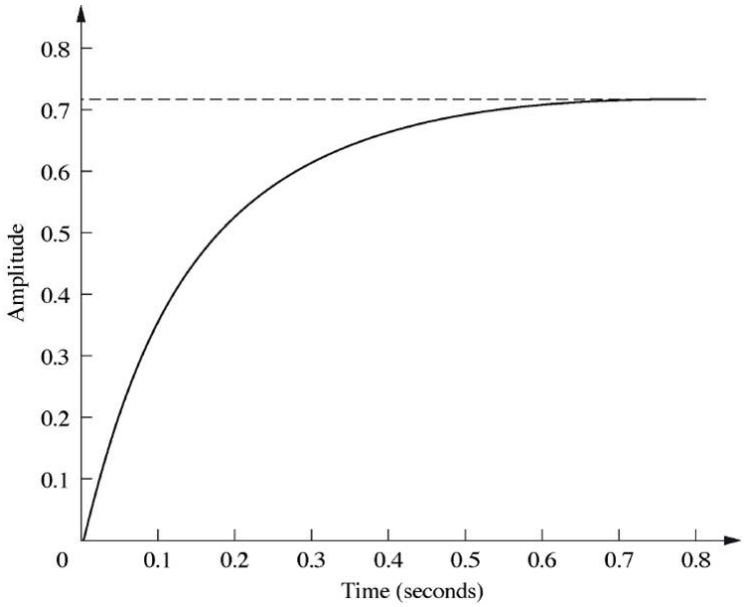


Figure: Real Pole

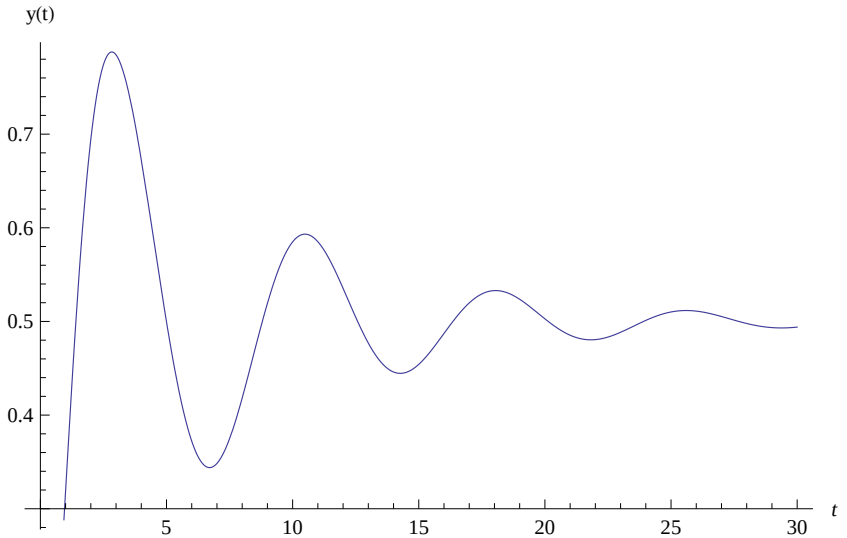


Figure: Complex Pair of Poles

We start with **Real Poles**

Step Response of a Real Pole

Rise Time and Settling Time

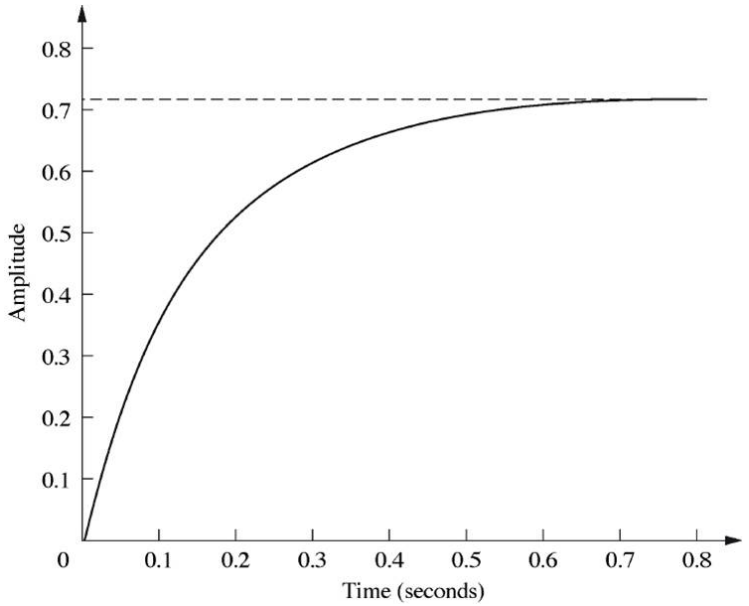
Consider the Step Response: $u(s) = \frac{1}{s}$.

The Solution: Use Partial Fraction Expansion

$$\hat{y}(s) = \frac{r}{s-p}\hat{u}(s) = \frac{r}{s-p}\frac{1}{s} = \frac{\frac{r}{p}}{s-p} - \frac{\frac{r}{p}}{s}$$

Inverse Laplace Transform:

$$y(t) = \frac{r}{p}\left(e^{pt} - 1\right)$$



Final Value:

$$y_{ss} = -\frac{r}{p}$$

Recall: For stability, $p < 0$!

Rise Time: How quickly does output reach final value over short times?
Settling Time: How quickly does output reach final value over long times?

Rise Time for a Real Pole (T_r)

Analytic Formula using step response

$$y(t) = \frac{r}{p}\left(e^{pt} - 1\right), \qquad y_{ss} = -\frac{r}{p}$$

Definition 6.

The rise time, T_r, is the time it takes to go from .1 to .9 of the final value.

If $y(t_1) = .1y_{ss} = -.1\frac{r}{p}$, then t_1 is:

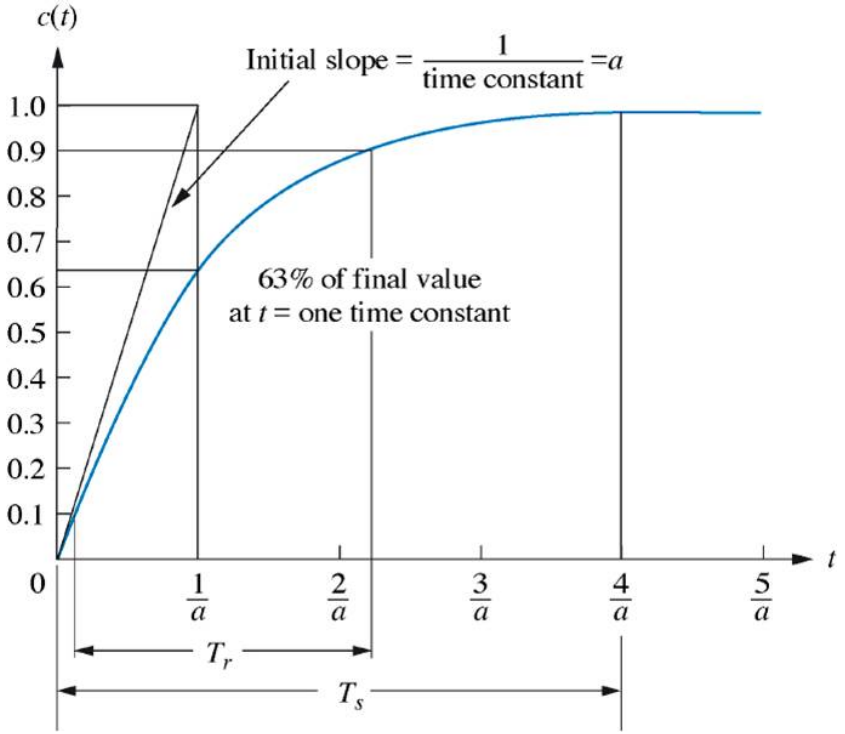
$$-.1 = e^{pt_1} - 1$$

$$\ln(1 - .1) = pt_1$$

$$t_1 = \frac{\ln .9}{p} = \frac{.11}{-p}$$

Likewise if $y(t_2) = -.9\frac{r}{p}$, then

$$t_2 = \frac{\ln .1}{p} = \frac{2.31}{-p}$$



Rise time (T_r) for a **Single Pole** is:

$$T_r = t_2 - t_1 = \frac{2.31}{-p} - \frac{.11}{-p} = \frac{2.2}{-p}$$

Definition 7.

The **Settling Time**, T_s, is the time it takes to reach and stay within .99 of the final value.

Settling Time: How quickly does output reach final value over long times?

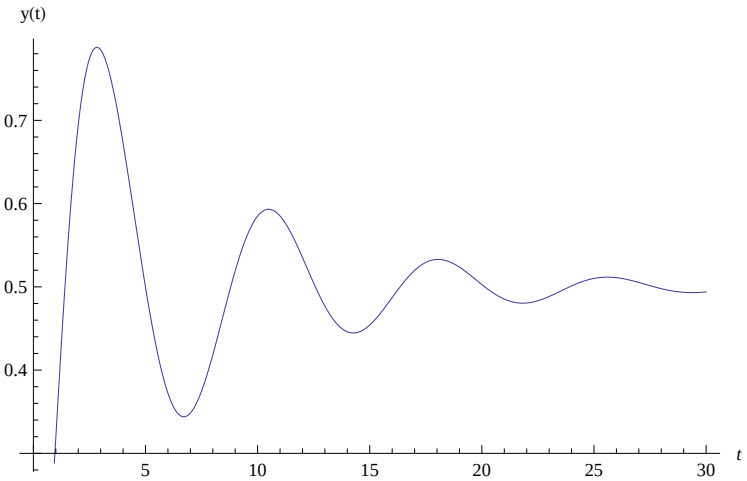


Figure: Step Response of Complex Poles

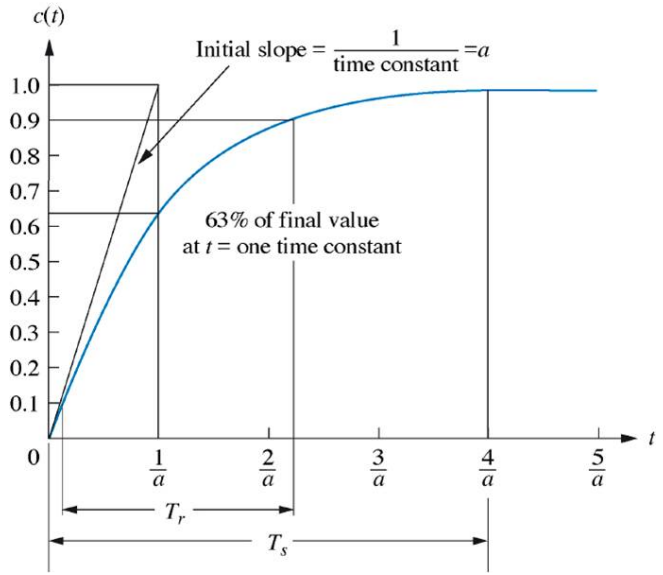


Figure: Step Response of Real Pole

LINK: Lots of Bouncing Balls!
LINK: Newton's Cradle

Settling Time for a Real Pole (T_s)

Analytic formula using step response

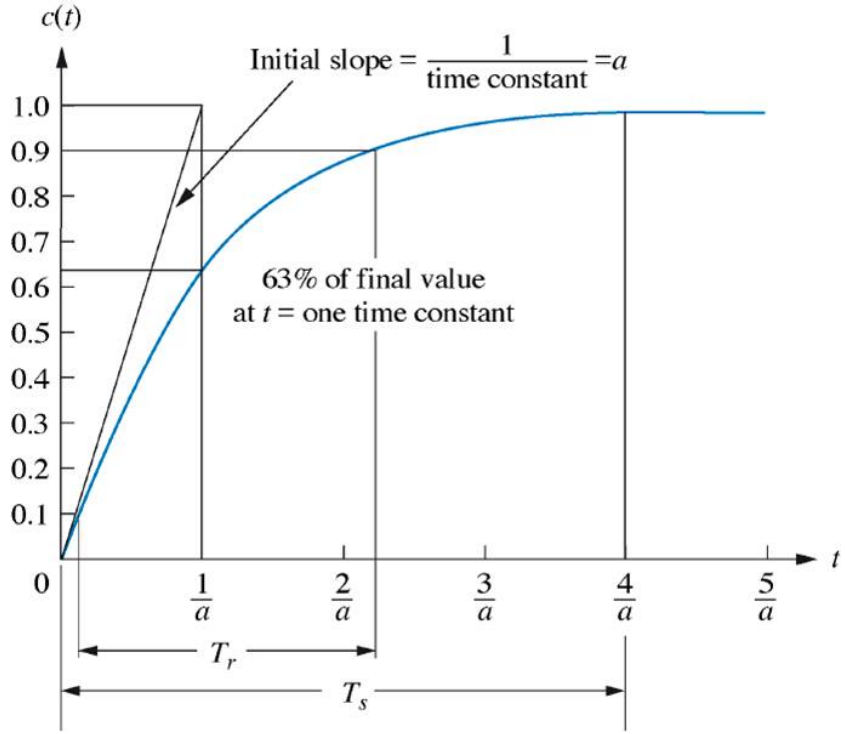
$$y(t) = \frac{r}{p}\left(e^{pt} - 1\right), \qquad y_{ss} = -\frac{r}{p}$$

Definition 8.

The **Settling Time**, T_s, is the time it takes to reach and stay within .99 of the final value.

Solve $y(T_s) = \frac{r}{p}\left(e^{pT_s} - 1\right) = -.99\frac{r}{p}$ for T_s:

$$-.99 = e^{pT_s} - 1$$
$$\ln(.01) = pT_s$$
$$T_s = \frac{\ln .01}{p} = -\frac{4.6}{p}$$



The settling time for a **Single Pole** is:

$$T_s = \frac{4.6}{-p}$$

Damped (ω_d) Frequency vs. Natural Frequency (ω_n)

Exponential Envelope for Complex Poles

Definition: Damped Frequency (ω_d) is the complex (imaginary) component of the pole location.

$$G(s) = \frac{\omega_d^2 + \sigma^2}{s^2 + 2\sigma s + \omega_d^2 + \sigma^2} = \frac{\sigma^2 + \omega_d^2}{(s+\sigma)^2 + \omega_d^2} = \frac{\sigma^2 + \omega_d^2}{(s + \sigma + \omega_d \imath)(s + \sigma - \omega_d \imath)}$$

Step Response:

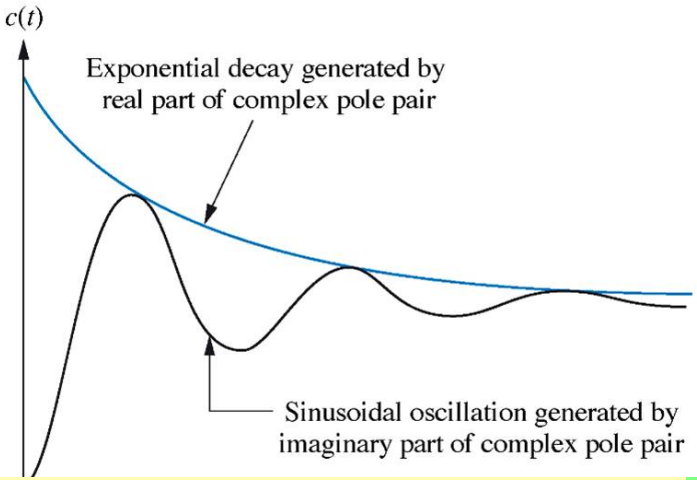
$$\hat{y}(s) = G(s)\frac{1}{s} = \frac{\sigma^2 + \omega_d^2}{(s+\sigma)^2 + \omega_d^2}\frac{1}{s} = -\frac{s + 2\sigma}{(s+\sigma)^2 + \omega_d^2} + \frac{1}{s}$$

Inverse Laplace Transform:

$$y(t) = 1 - e^{\sigma t}\left(\cos(\omega_d t) + \frac{\sigma}{\omega_d}\sin(\omega_d t)\right)$$

The result is oscillation with an Exponential Envelope.

- Envelope decays at rate σ
- Speed of oscillation is ω_d, the **Damped Frequency**



Damped (ω_d) Frequency vs. Natural Frequency (ω_n)

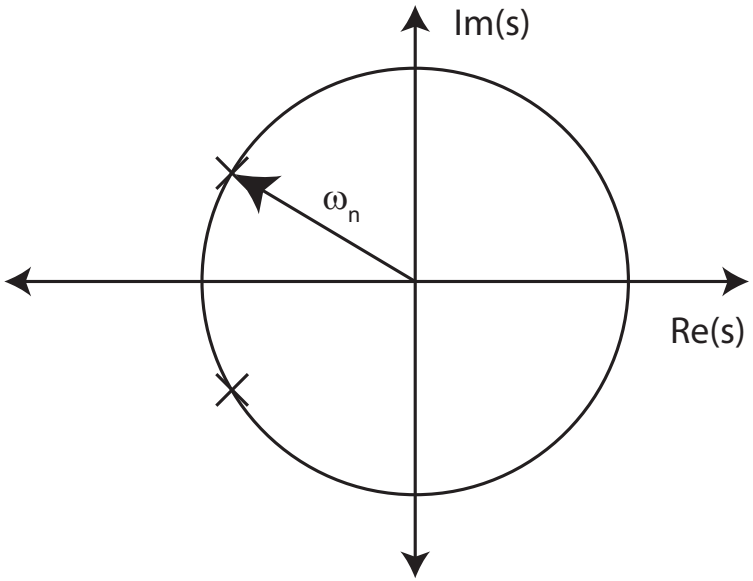
Exponential Envelope for Complex Poles

Definition: Natural Frequency (ω_n) is the *Magnitude* of the pole location $|p|$.

$$G(s) = \frac{\omega_n^2}{s^2 + 2\omega_n\zeta s + \omega_n^2}$$

Definition 9.

The **Natural Frequency** of a pole at $p = \sigma + \imath\omega_d$ is $\omega_n = \sqrt{\sigma^2 + \omega_d^2}$.



- for $\hat{y}(s) = \frac{1}{s^2+as+b}\frac{1}{s}$, $\omega_n = \sqrt{b}$.
- Radius of the pole in complex plane.

AKA: Resonant Frequency

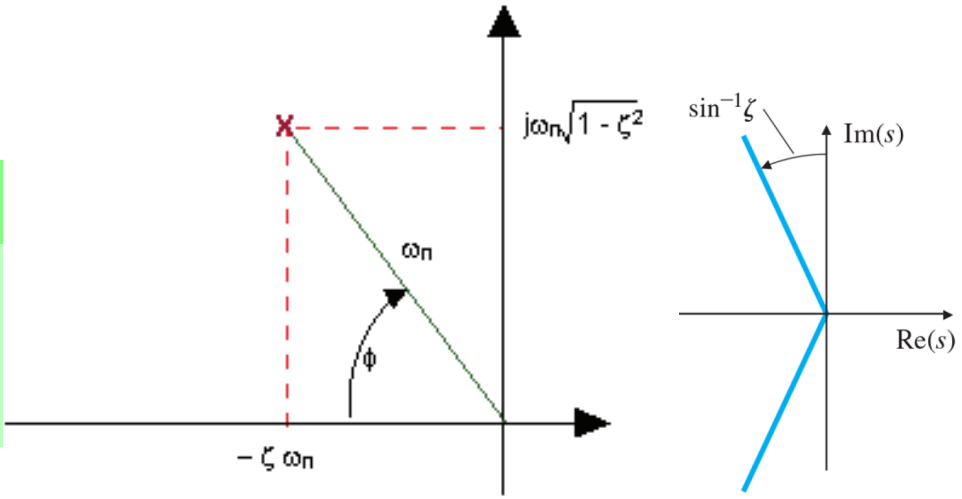
- Input at frequency ω_n has highest steady-state amplification.
- $\max_{\omega\in\mathbb{R}} |G(\imath\omega)| = |G(\imath\omega_n)| = |\frac{1}{2\zeta}|$

Damping Ratio for Complex Poles

Damping ratio is often more important than σ.

Definition 10.

The **Damping Ratio** of a pole at $p = \sigma + \imath\omega_d$ is $\zeta = \frac{|\sigma|}{\omega_n} = \frac{|\sigma|}{\sqrt{\sigma^2+\omega_d^2}}$.



- for $\hat{G}(s) = \frac{1}{s^2+as+b}$, $\zeta = \frac{a}{2\sqrt{b}}$.
- Gives the ratio by which the amplitude decreases per oscillation (almost...).

Recall: At Resonant Frequency

- Input: $u(t) = \sin(\omega t)$
- Gives Output (Bode Plots): $y(t) = |G(\imath\omega)|\sin(\omega t + \angle(G(\imath\omega)))$
- and $|G(\imath\omega_n)| = |\frac{1}{2\zeta}|$

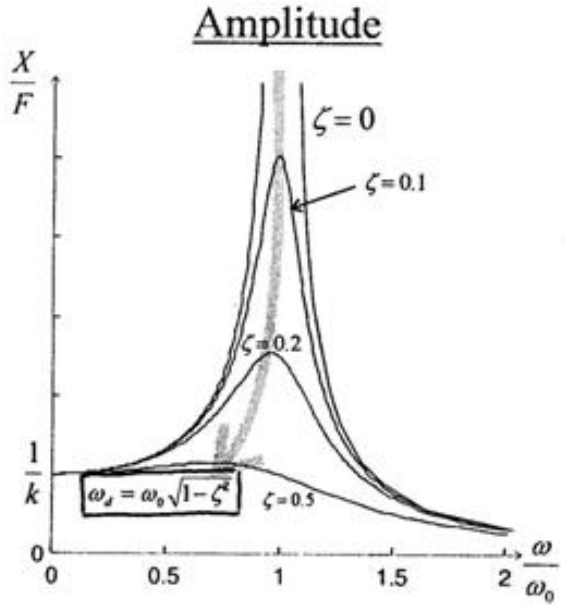
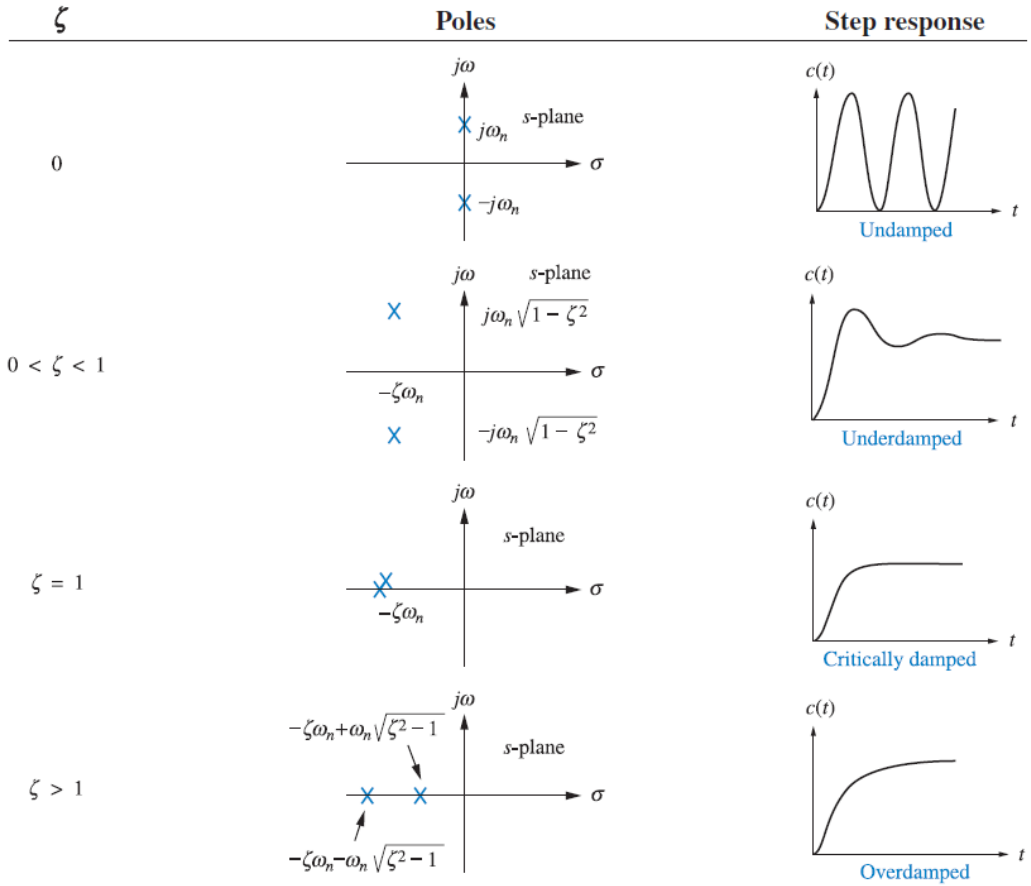


Figure: $|G(\imath\frac{\omega}{\omega_n})|$ vs. ζ

Damping Ratio and Pole Locations

We use several adjectives to describe damping ratio, ζ:

- Undamped
  - Oscillation continues forever, $\sigma = \zeta = 0$. Imaginary poles. $p = \pm \imath\omega$
- Underdamped
  - Oscillation continues for many cycles. $\zeta < 1$. Comples poles. $p = \sigma \pm \imath\omega_d$
- Critically Damped
  - No oscillation or overshoot. $\zeta = 1$. Repeated real poles. $p_1 = p_2$
- Overdamped
  - When $\zeta > 1$, poles are real. $p_1 = -\sigma_1$, $p_2 = -\sigma_2$.



Summary

What have we learned today?

Characteristics of the Response

Real Poles

- Steady-State Error
- Rise Time
- Settling Time

Complex Poles

- Complex Pole Locations
- Damped/Natural Frequency
- Damping and Damping Ratio

Continued in Next Lecture